



Mom, can you help me understand the Repeat Until block in Scratch?

Sure, this block keeps repeating an action until a condition is true. Just like a school bell or a timer.

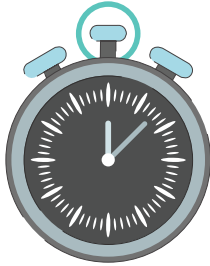
Can you show me how it works?

Yes, let me show you by creating a timer.



Scratch is a software that uses a block-based code. Blocks are arranged in a particular order to give instructions. In this lesson, we will learn to create our own timer in Scratch and use a fun theme to make it attractive.

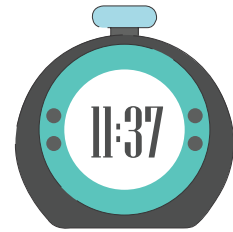
Timers come in handy to keep track of time when we do some tasks. For example, we can use a timer to track time when we work, when we play games like scrabble or chess, or when we cook or bake.



Clock timer



Digital timer



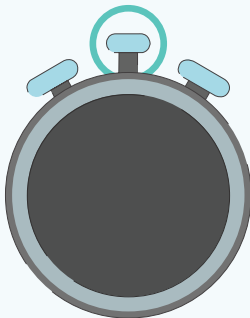
Watch timer

Different types of timers

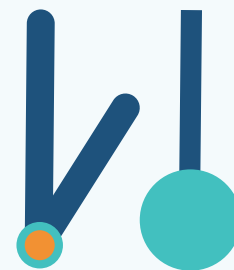
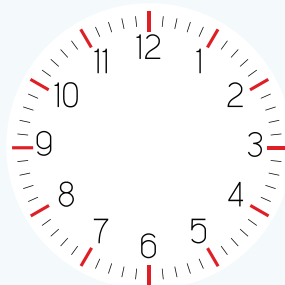
Activity

In this activity, we will create a clock shaped timer. For this, we need the following:

- For the clock face: A background with numbers to tell the time
- Clock hands



Clock faces

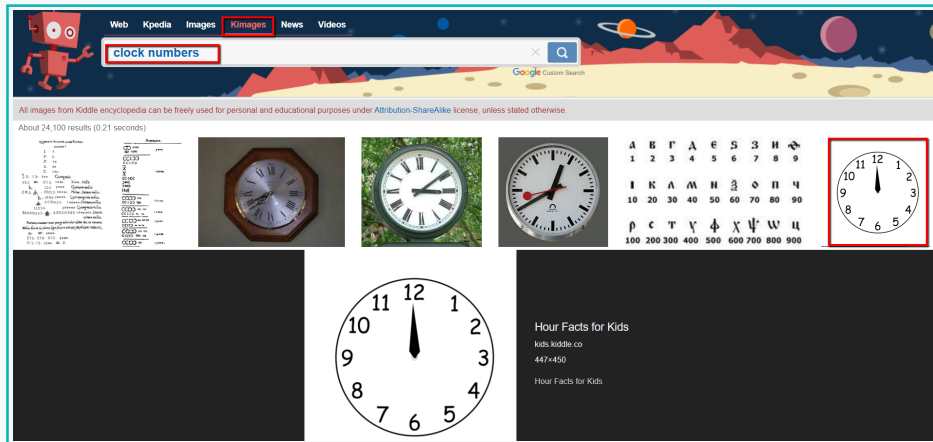


Clock hands

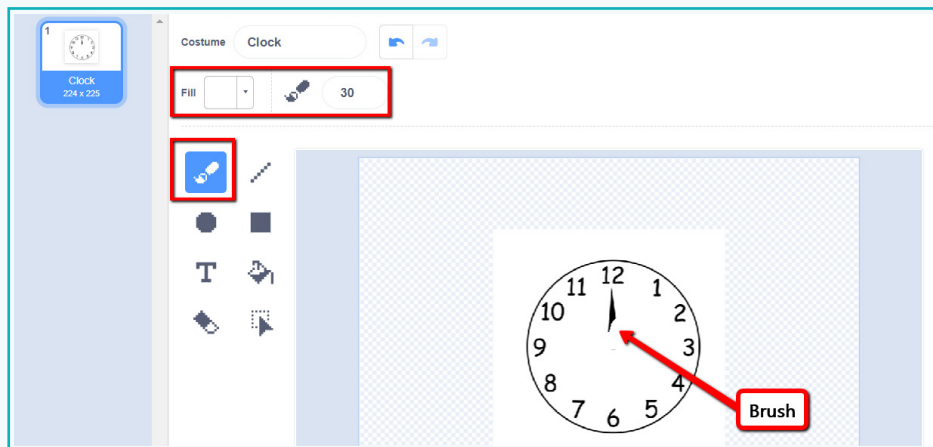
Activity

Step 1: Create a backdrop of a clock face for the clock timer

- Open Scratch.
- Select the 'clock face' backdrop from the Scratch library.
- If it is not available, find and download an image from Kiddle.
 - Go to Kimages on Kiddle and search for "clock numbers".
 - Select the clock face background shown and download the picture.



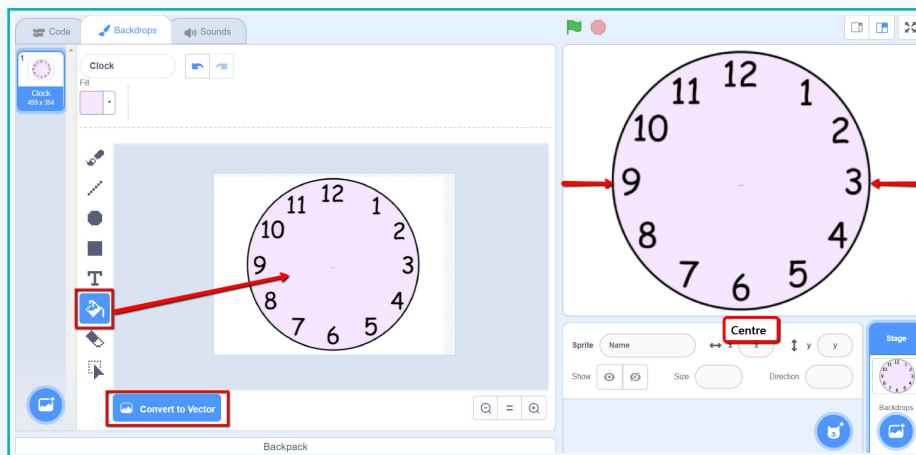
- Open Scratch and delete the default sprite.
- For the clock background, upload the image of the clock face as the backdrop.
- We only need the clock face for the backdrop. Use the Brush tool in white colour to erase the clock hand.



Activity

- g. Use the 'Fill' tool to change the background colour.
- h. To change the size of the clock, click on 'Convert to Vector'.
- i. Resize the clock to fit the stage.

Note: Place the backdrop such that centre of the clock is at the centre of the stage.

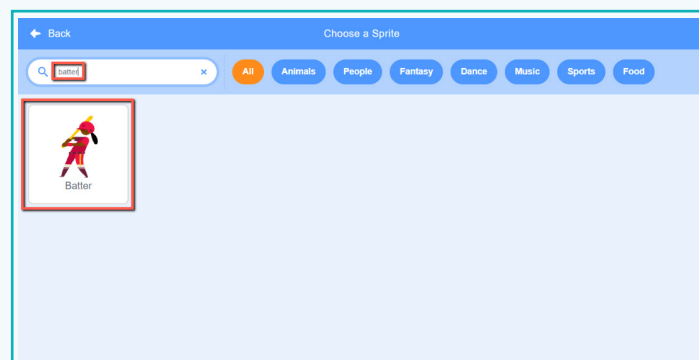


The clock timer we want to create will only have a 'second hand'. We don't need a minute and hour hands.

Step 2: Create a 'second hand' for the clock timer

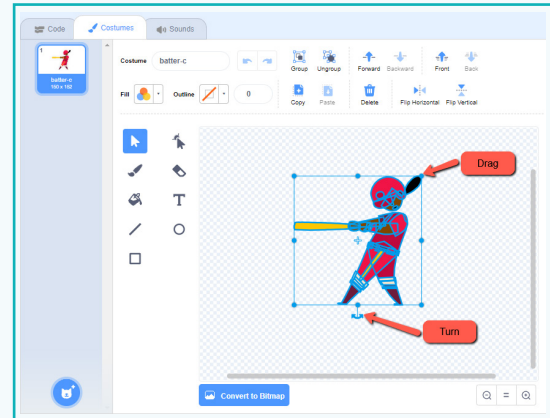
To make our timer look unique, we can choose a theme for it. Our theme for this one will be **sports**.

- a. Click on 'Choose a Sprite' and search for 'batter'. Select the 'Batter' sprite.



Activity

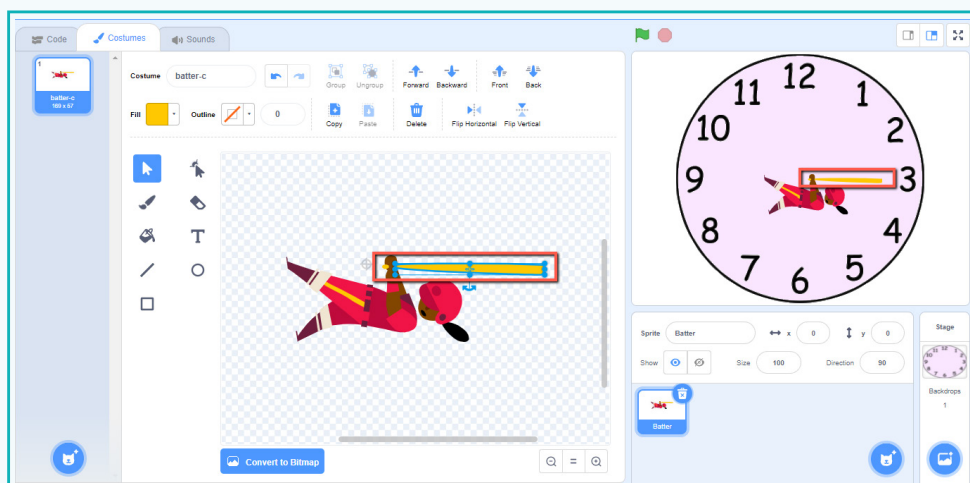
- b. Except the 'batter-c' costumes, delete the remaining costumes.
- c. To select the entire sprite in Costumes, press keys 'Ctrl + A' together.
- d. Reduce the size of the sprite.



- e. Click on the batter's bat, and rotate and drag to increase its length. The 'second hand' should be long enough to reach the numbers on the clock.

The default or original position of all sprites in Scratch is towards the right. In our timer, the batter's bat is the 'second hand' on the clock. The bat should point towards the right at the beginning.

- f. Turn the image to make the bat horizontal and point it towards the right. It should point at 3 o'clock on the stage.



The 'second hand' is now ready to move. Now time for some action!

Exercise

Tick mark ☒ the correct answer

1 When pressed together, which keys select the entire sprite?

a. Ctrl + A ☐

b. Ctrl + B ☐

c. Ctrl + C ☐

Activity

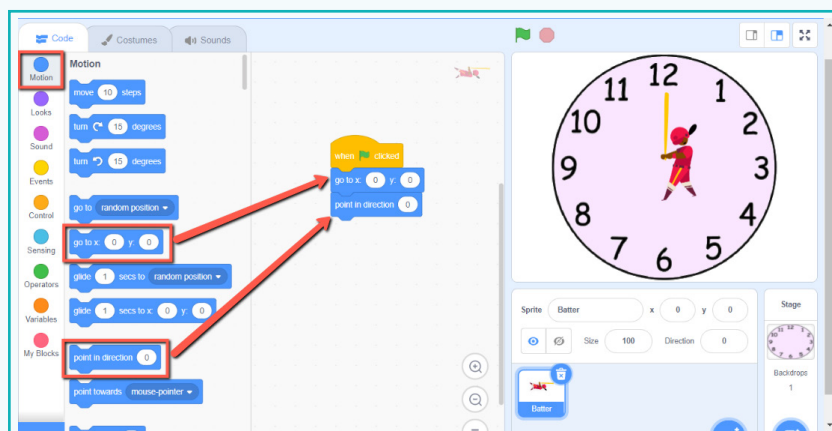
In a timer, time in seconds is taken as an input to start the clock. We set a number of seconds to be completed. Once they have passed, the alarm rings and time is up.

Step 3: Make the 'second hand' move

Now that our clock timer is ready, we need the second hand to move.

a. The timer should start when  is clicked. Add the  block.

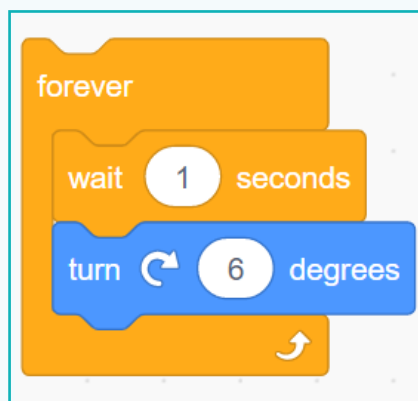
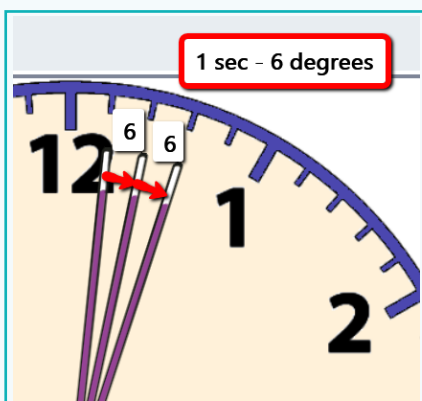
b. Add the 'go to x y' and 'point in direction' blocks to position the 'second hand' at the centre of the stage. It should point upwards at 12 o'clock.



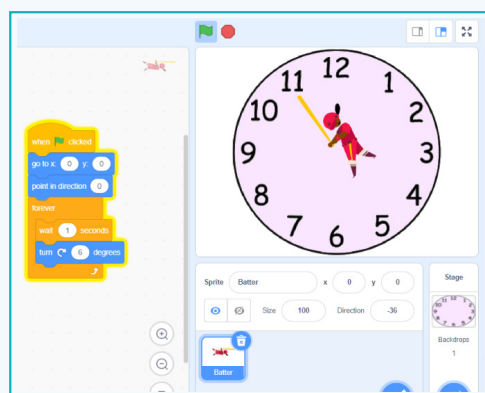
The 'second hand' moves every second. It rotates 6 degrees in each second.

Activity

- c. The 'second hand' needs to continuously rotate. To make this happen, add the 'forever' block.
- d. Insert blocks to turn the hand 6 degrees to the right after each second. Add the 'forever' block to the code.



- e. When  is clicked, the 'second hand' will start to move like a real clock.



To tell the time, we need to count the number of seconds that have passed.

- f. 60 seconds = 1 minute.

Time continuously moves. We can store a number to represent the current time. We can set it to update as the seconds pass. For this, we can use **Variables**.

Variable: A variable is a changeable value recorded in Scratch memory. It is a storage place with a name that contains information in the form of numbers or words.

Variables have two parts:

- The name of the variable
- The value in the variable

For example: Madhu puts her pocket money into her piggybank.

Activity

To keep track of her money, she uses a variable called “**myMoney**”.

myMoney

She sets variable value to the current amount she has. **myMoney = 40.**

set myMoney ▼ to 40

The next day, **grandma gives her Rs. 20. She adds the amount in the myMoney variable to keep track of it.**

change myMoney ▼ by 20

After she **changes myMoney by 20**, the amount in the **variable becomes 60.**

change myMoney ▼ by 20



If we use a variable, we don't need to change the value of the sprite to change the time.

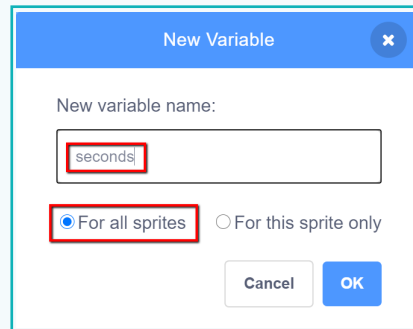
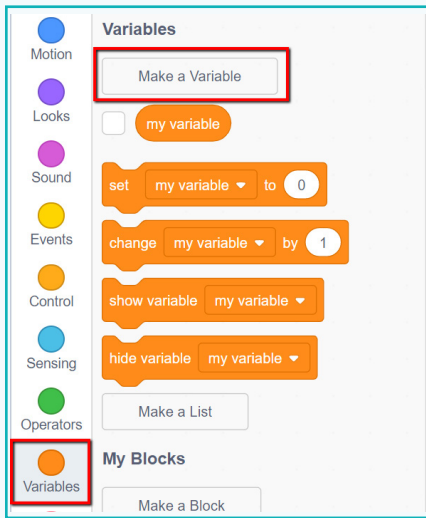
Let's learn how to do this:

Step 4: Add a variable to measure time

- Go to the 'Variable' palette and click on 'Make a Variable'.
- Create a variable and name it 'seconds'.

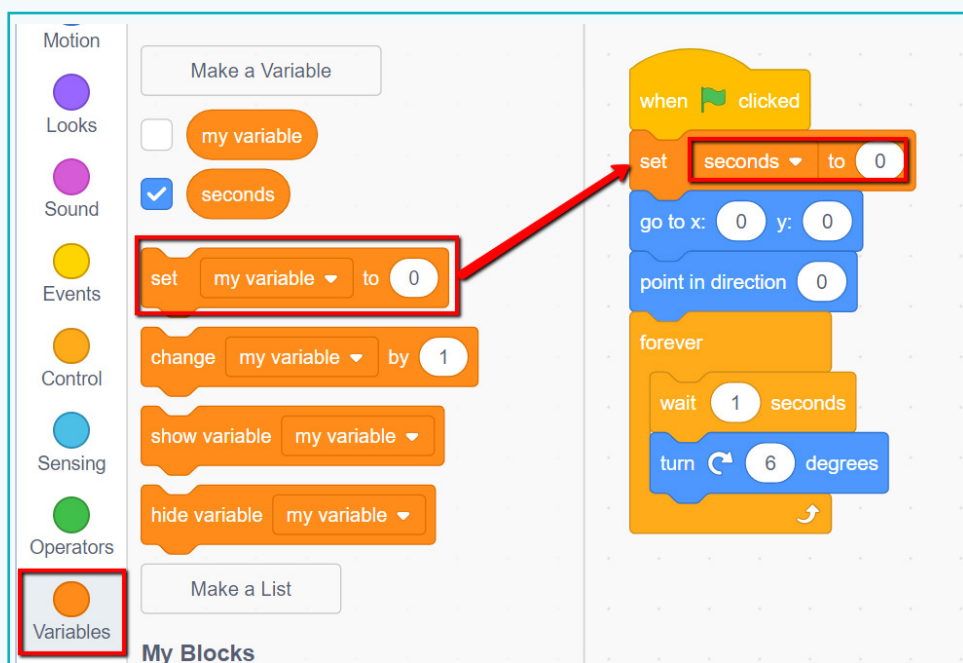
Activity

c. Select the option 'For all sprites'.



A variable **seconds** will appear in the palette to store the number of seconds that have passed.

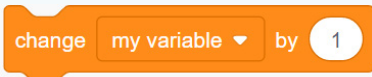
d. Use the 'set variable to' block to set the initial value of the variable to '0'.



Activity

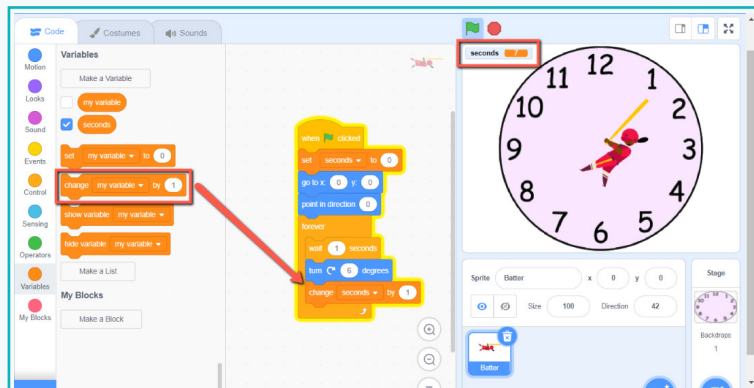


This block sets the selected variable to a given value. The value can be a number or a word.



This block changes the value in specified variable.

e. Add the 'change seconds by 1' block to change the variable value at each second.



We now have a working clock! The clock records time as it passes in seconds.

Exercise

☒ Tick mark ☒ the correct answer

2 Which block is used to set the initial value in a variable?

- a. ☐ set my variable to 0 b. ☐ change my variable by 1 c. ☐ show variable my variable

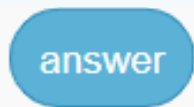
Activity

To take input to set the timer, we need to add code. The timer needs to be set before the clock starts to tick.

Step 5: Add code to take input

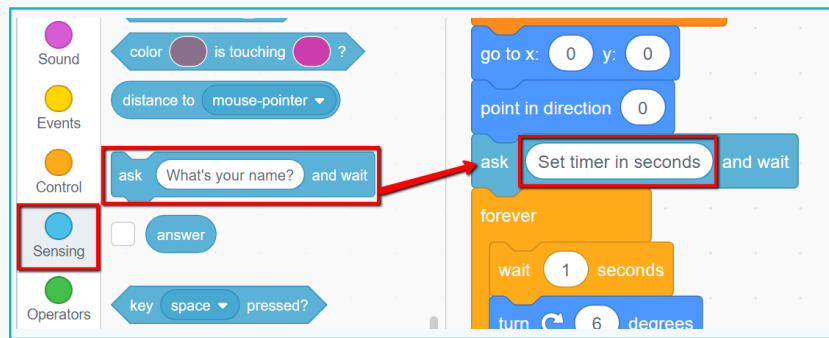


This block instructs the sprite to ask a question. An input box is displayed at the bottom of the screen for the user to type an answer.



This is a variable block that holds the most recent text added as an answer to the 'Ask () and Wait' block. When there is no input, the value will be nil.

a. Add the 'ask and wait' block. It should ask to set the timer in seconds.



Exercise

Tick mark ☒ the correct answer

3 If 1 minute = 60 seconds and 1 hour = 60 minutes, how many seconds make 1 hour?

a. 3600 ☐

b. 60 ☐

c. 1000 ☐

Activity

Currently, the clock timer is set to repeat forever. However, we want the clock to run only for the duration that is set in the timer. For this, we use the **Repeat Until** block.

Repeat Until: This block repeats the code inside it until the condition for the block is fulfilled. It is a combination of the repeat and conditions blocks and is found inside the 'Control' palette.

For example: Hari's family has a good habit; they do not waste food. They continue to eat (repeat) until they finish the food in their plates.

There are two parts in this action:

- a. **Action to be repeated:** Eating
- b. **Repeat until this condition is satisfied:**
Finish food on the plate.



Bytes

If the number of seconds passed are more than 60, time can be represented in terms of minutes and seconds.

For example:

1 minute, 2 seconds = 1 minute + 2 seconds

As 60 seconds = 1 minute

So, = 60 seconds + 2 seconds
= 62 seconds

Similarly, 2 minutes 15 seconds = 1 minute + 1 minute + 15 seconds
= 60 seconds + 60 seconds + 15 seconds
= 135 seconds

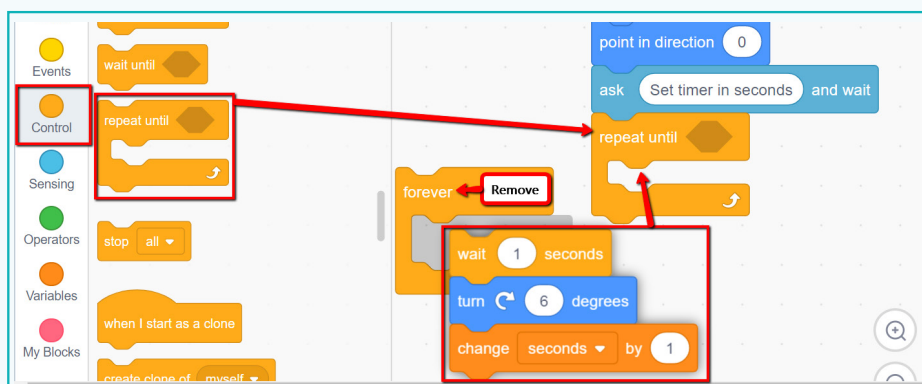
This helps us always set the time value in seconds.

Activity

In our case, the clock should run only for the time set in the timer and not forever.

Step 6: Make the clock run for the time duration set in the timer

- Remove the 'forever' block and replace it with the 'repeat until' block.
- All the code blocks inside the 'forever' block will be shifted to 'the repeat until' block.



Now that we have added the 'repeat until' block, we need to add the condition. For this, we use **Operators**.

Operators: An Operator block is a colour-coded light-green block used to perform logical comparisons, math calculations, and modify words or sentences.

The **comparison operator** block compares values and answers either true or false. Variables can also be used in these blocks.

For example:



This block checks if the value of the left-hand side is greater than the right-hand side.
Here 100 is greater than 50, so the result will be true.



This block checks if the value of the left-hand side is greater than the right-hand side.
Here 100 is greater than 50, so the result will be false .

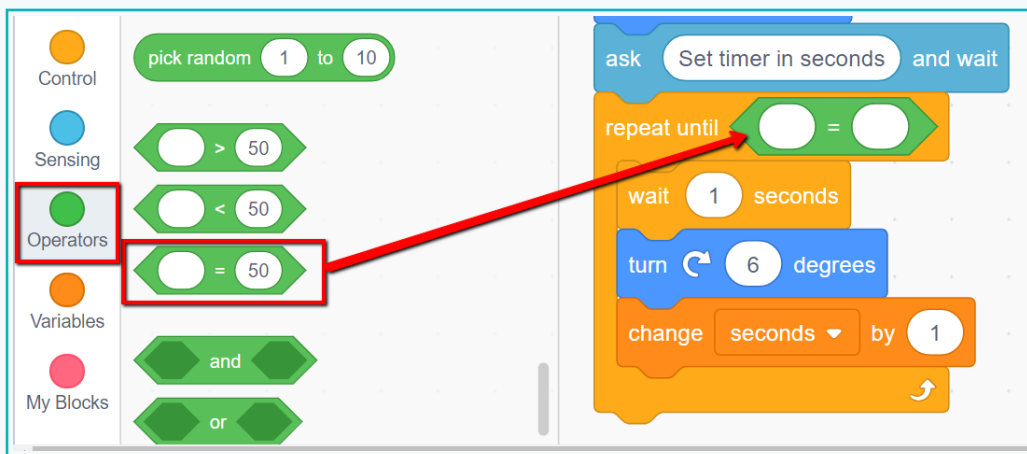


This block checks if both sides have the same value. If the value in the 'y' variable is 50, the result will be true. If it is not, the result will be false.

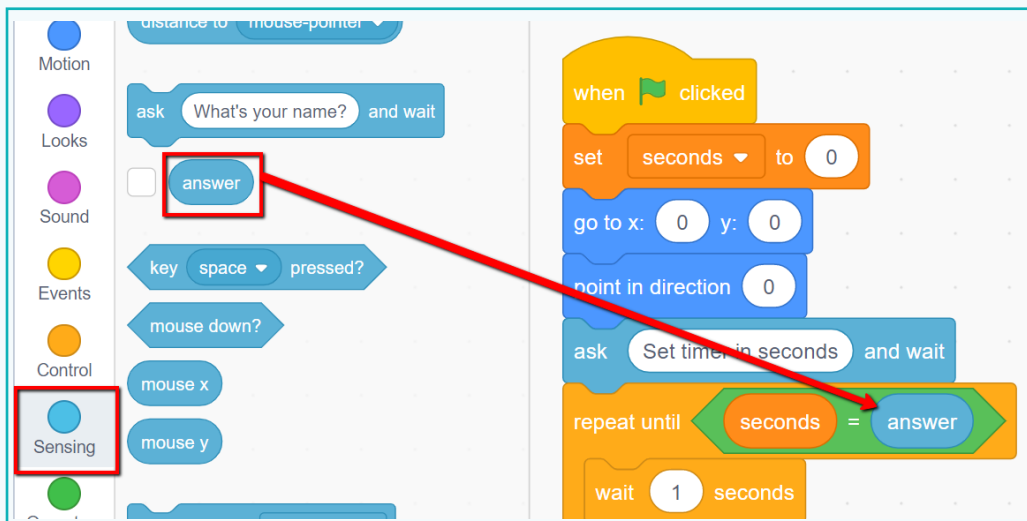
Activity

Step 7: To add a condition in the Repeat Until block

a. Add an 'operator' block to compare if 'seconds = answer'. This will contain the time set in the timer.



b. Add the 'seconds' variable on the left-hand side and the answer on the right-hand side of the '=' symbol.



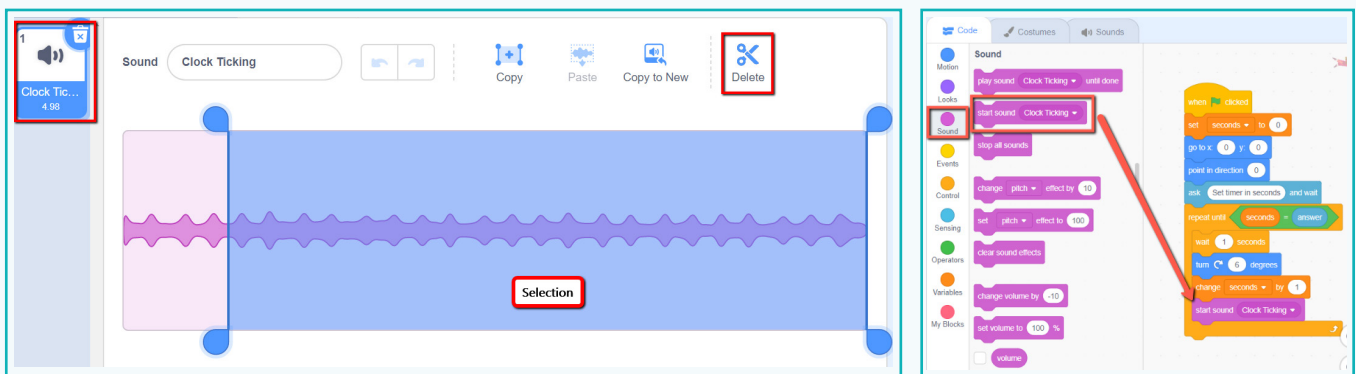
When we click , the 'second hand' will run for the time set in the timer. We can set a sound to show the clock ticking every second. When the time is up, we can set a different sound like an alarm clock.

Activity

Step 8: Add sound and visual effects

Sound: A sound is an item which can be heard. It can be played in a Scratch project, available in the application's built-in sound library. The Sound blocks are used to play sounds.

- Add a 'clock ticking' sound from the library.
- We need only one ticking sound for each second. For this, we need to reduce the length of the sound.
- Add the 'start sound' block in the 'repeat until' block. This will add the 'clock ticking' sound to each second.

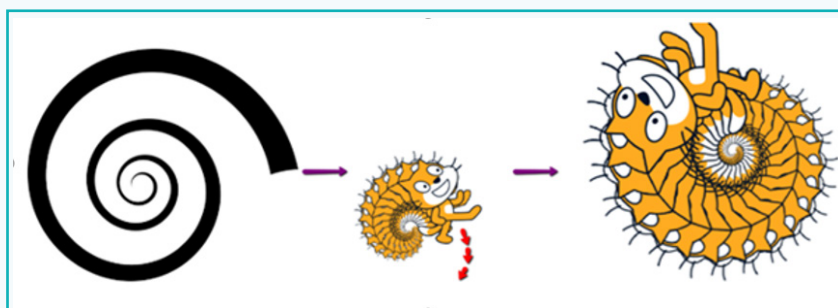


When time is up, we can make the batter appear to move towards us. For this, we will use a visual effect.

Visual effect: A visual effect is used on a sprite to change its appearance. It can make imaginary things seem real.

Scratch has many different effects. Here, we will use the spiral effect since the 'second hand' moves in circles.

Spiral movement is a continuous circular movement. It gradually moves away from the centre as it revolves around it.



Activity

We will add a spiral effect to the 'batter' sprite.

d. Add the 'change size by' block and set its value to **20**.

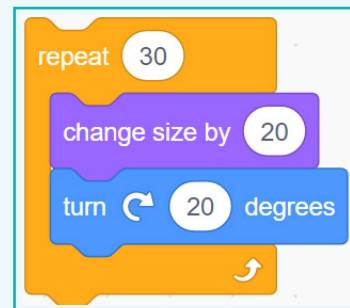
This will increase the size of the sprite to create the effect of it moving towards the user.

e. Add the 'turn by' block and set its value to **20** degrees. This will give us the spiral effect that we want.

f. Insert the above blocks into the 'repeat' block and set it to repeat **30** times.

g. Add this block after the 'repeat until' block.

When time is up, the batter will move according to the code and rotate in a spiral.



Step 9: Notify the user when time is up

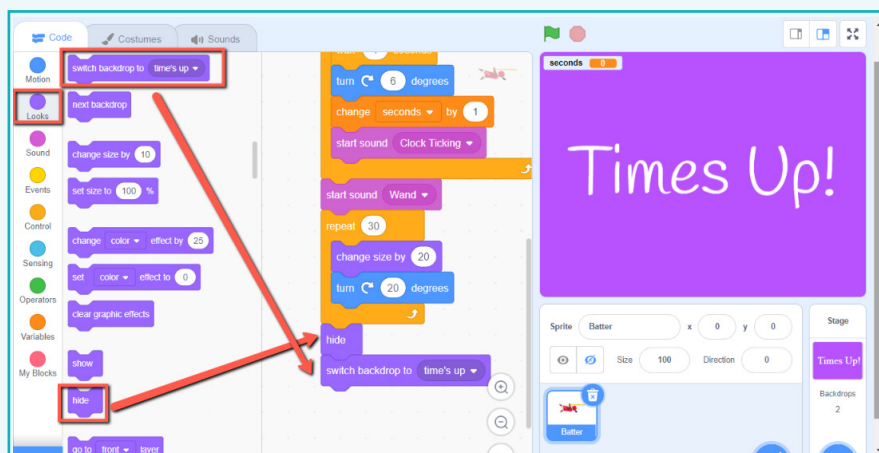
We need to create a backdrop with the text 'Time's Up!'

a. Use the 'Hide' block to hide the sprite.

b. To show the 'Time's Up' block, add the 'switch backdrop to' block.

c. Add the sound called 'Wand' from the library.

d. To indicate that time is up, add the 'start sound' block after repeat until and before visual effect starts.



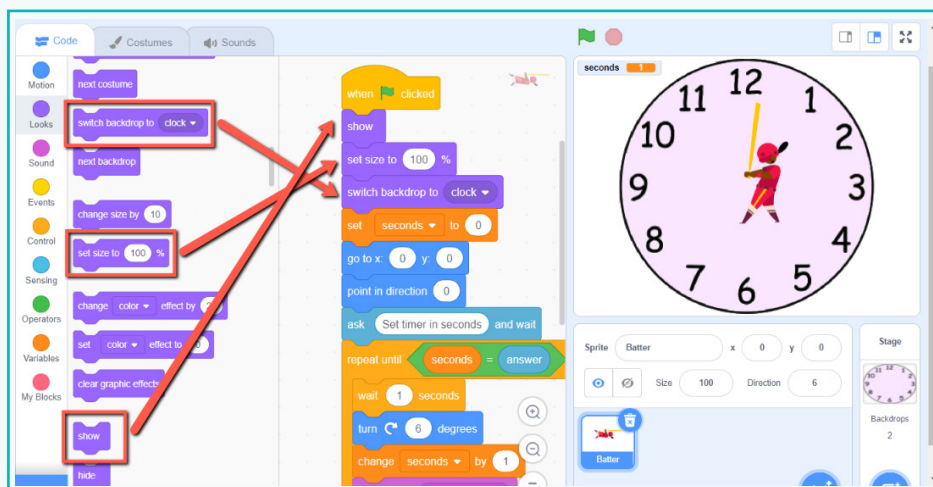
Activity

The effect looks great! However, if we start the timer again, we need to reset the 'clock' backdrop and the sprite back to its original size.

e. Add the 'show' block after the  block at the beginning of the code.

f. Add the 'set size to' block and set its value to '100%'.

g. Add the 'switch backdrop to' block to set the clock background. This will start the timer again.



Our sports themed clock timer is ready!

Scratch lets us create our own interactive stories, games, and animations. A variable in Scratch is a changeable value recorded in Scratch's memory. It contains information in the form of numbers or words. The 'Repeat Until' block repeats the code inside it until the condition is fulfilled. We can also create visual effects in Scratch using code to enhance the look of animations.

Exercise

 Tick mark ☒ the correct answer

4 Which operator block is used to compare if two values are equal?

- a. ☐  b. ☐  c. ☐ 

 State whether True or False

5 The 'Repeat until' block is used to repeat a set code until a specific condition is satisfied.

 _____

6 A variable can store numbers as well as words.

 _____

7 The name of the variable tells us what the value inside the variable stands for.

 _____

Now I can use this timer while playing Scrabble with my friends.

Certainly!



Tech Flash

- **Variables** : A variable is a storage place with a name that contains information in the form of numbers or words.
- **Repeat until** : This block repeats code inside it until the condition for the block is fulfilled.
- **Comparison operators** : This operator block compares values on the left-hand side and right-hand side to decide which is greater than, less than or equal.